KAUSHIK ROY

Assistant Professor of Computer Science | The University of Alabama | kroy2@ua.edu | 3031 Cyber Hall

Research

Neurosymbolic AI; process-grounded learning; causal reasoning; trustworthy agents; explainable decisions; healthcare and manufacturing.

Appointments and affiliations

2025–present: Assistant Professor, University of Alabama

Research affiliate: BRAINS Lab, University of Alabama

2020–2025: Graduate Research Assistant, Artificial Intelligence Institute, University of South Carolina

Education

Ph.D. Computer Science and Engineering, University of South Carolina, 2025; advisor Amit Sheth; dissertation Process-Grounded Knowledge-Infused Learning and Decision Making

M.S. Computer Science, Indiana University Bloomington, 2017

B.E. Computer Science, R.V. College of Engineering, 2015

Research experience

Bosch Center for Artificial Intelligence: summer 2023 and summer 2024

Samsung Research America: summer 2022

The Alan Turing Institute: fall 2021

StARLinG Lab, University of Texas at Dallas: 2017–2020

Research support and proposal activity

2026: collaborator, UA SCoE CORE Equipment Funding, multi-investigator award ($30,000 total; no individual allocation claimed)

Contributions to NSF EAGER #2335967 and #2133842 (PI Amit Sheth)

2026: proposals and concepts in trustworthy AI, manufacturing data and causal decision support; not claimed as awarded funding

Publications — current verified website bibliography

• Exploring the Potential of Large Language Models for Assisting with Mental Health Diagnostic Assessments Kaushik Roy , Harshul Surana, Darssan Eswaramoorthi, Yuxin Zi, Vedant Palit, Ritvik Garimella, Amit Sheth ACM Transactions on Computing for Healthcare · 2026 · Peer-reviewed journal DOI / Paper ■

• How Compliant is Sepsis Treatment? An Expert-Guided Neuro-symbolic Pipeline for Generating Clinical Compliance Insights Himanshu Tripathi, Kaushik Roy , Subash Neupane, Shahram Rahimi NeSy 2026 · Accepted conference paper; arXiv preprint arXiv ■ Conference ■

• In-Situ Eval: A Modular Framework for Custom and Real-Time RAG Benchmarking Ritvik Garimella, Kaushik Roy , Chathurangi Shyalika, Amit Sheth Proceedings of AAAI · Demonstration Track · 2026 DOI / Paper ■

• Metatuning: An Empirical Study of Judge-Guided Prompt Refinement and Its Boundary Conditions Aniruddha Chattopadhyay, Raj Dandekar, Kaushik Roy Neurosymbolic Artificial Intelligence · 2026 · Journal DOI / Paper ■

• Learning and Reasoning with Model-Grounded Symbolic Artificial Intelligence Systems Aniruddha Chattopadhyay, Raj Dandekar, Kaushik Roy Proceedings of the 19th International Conference on Neurosymbolic Learning and Reasoning (NeSy) · PMLR 284:957–976 · 2025 Paper ■

• Cognitive Neurosymbolic Artificial Intelligence for Complex Decision-Making: Integrating Foundation Models, Cognitive Architectures, and Knowledge Yuxin Zi, Kaushik Roy , Amit Sheth IEEE Intelligent Systems · 40(3):16–22 · 2025 DOI / Paper ■

• NeuroSymbolic Knowledge-Grounded Planning and Reasoning in Artificial Intelligence Systems Amit Sheth, Vedant Khandelwal, Kaushik Roy , Vishal Pallagani, Megha Chakraborty IEEE Intelligent Systems · 40(2):27–34 · 2025 DOI / Paper ■

• Composite AI With Custom, Compact, Neurosymbolic Models: The Emergent Enterprise Artificial Intelligence Paradigm Amit P. Sheth, Kaushik Roy , Revathy Venkataramanan, Venkatesan Nadimuthu, Chathurangi Shyalika IEEE Internet Computing · 29(2):37–49 · 2025 DOI / Paper ■

• Causal Event Graph-Guided Language-based Spatiotemporal Question Answering Kaushik Roy , Alessandro Oltramari, Yuxin Zi, Chathurangi Shyalika, Vignesh Narayanan, Amit Sheth AAAI Symposium Series · 2024 Paper ■ Code ■

• RI2AP: Robust and Interpretable 2D Anomaly Prediction in Assembly Pipelines Chathurangi Shyalika, Kaushik Roy , Renjith Prasad, et al. Sensors · 2024 DOI / Paper ■

• Healthcare Assistance Challenges-Driven Neurosymbolic AI Kaushik Roy Biomedical Journal of Scientific & Technical Research · 2024 DOI / Paper ■

Additional research outputs

• Ramnani, Roy and Sheth (2026). MATRIX: Mental heAlth diagnostics Through Real time Intelligent unified X-AI attribution reasoning. Frontiers in Digital Health.

• Pallagani, Roy et al. (2024). On the Prospects of Incorporating Large Language Models in Automated Planning and Scheduling. ICAPS.

• Roy, Khandelwal et al. (2024). GEAR-Up: Generative AI and External Knowledge-Based Retrieval. AAAI.

• Roy, Gaur et al. (2023). Proknow: Process Knowledge for Safety Constrained and Explainable Question Generation. Frontiers in Big Data.

• Roy, Zhang, Gaur and Sheth (2021). Knowledge-infused Policy Gradients with Upper Confidence Bound for Relational Bandits. ECML PKDD.

Selected invited talks and service

• ISWC 2024 tutorial: Neurosymbolic Customized and Compact CoPilots

- Ontolog Summit 2024: Healthcare Assistance Challenges-Driven Neurosymbolic AI
- University of Florida ICCC 2024 invited talk
- Guest Editor: IEEE Internet Computing and Information special issues
- Program Committee: AAAI 2024 Fall Symposium on LLMs for KG and Ontology Engineering
- Organizing Committee: KiL 2023 at KDD; CASY 2022

Mentoring and outreach

Ph.D. student: Ashraf Abdul-Muumin. Curriculum development: CS 483/583 Computational Foundations of Machine Learning.

Film: AI in the Family (2025), IMDb tt38192436. Personal YouTube: youtube.com/@kaushikroy4041.

Verification

UA directory: eng.ua.edu/eng-directory/dr-kaushik-roy/ | ORCID: 0000-0001-6610-7845 | Scholar Commons dissertation 8563

Prepared October 2026 from newly curated public records and current publication entries; this is a selected CV.